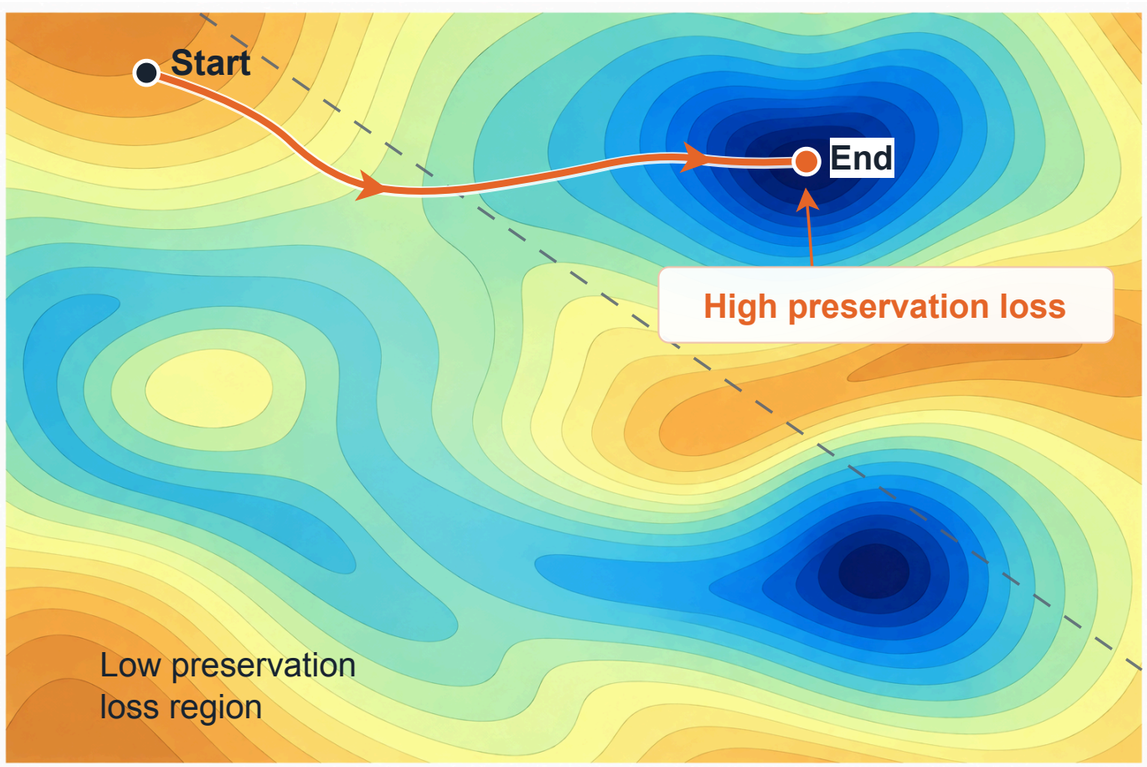
Downstream task loss Low High Low-curvature delta space

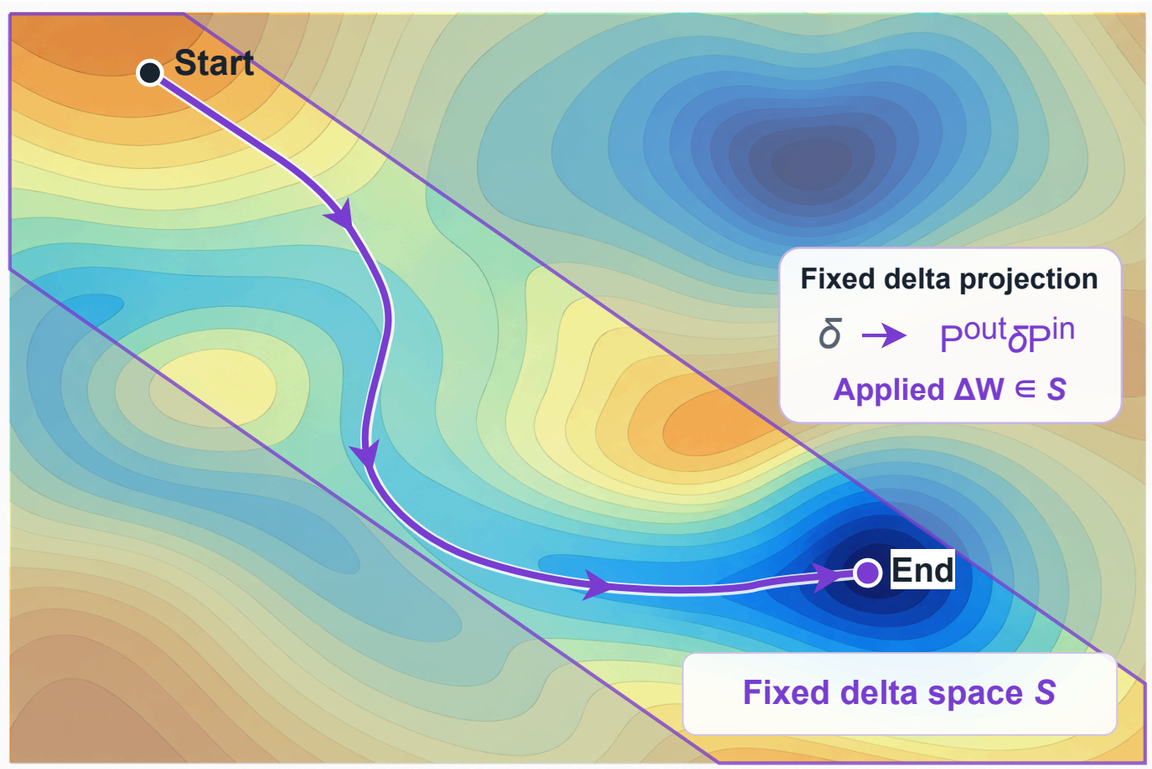
(a) Unconstrained optimization

Low task loss · high preservation loss

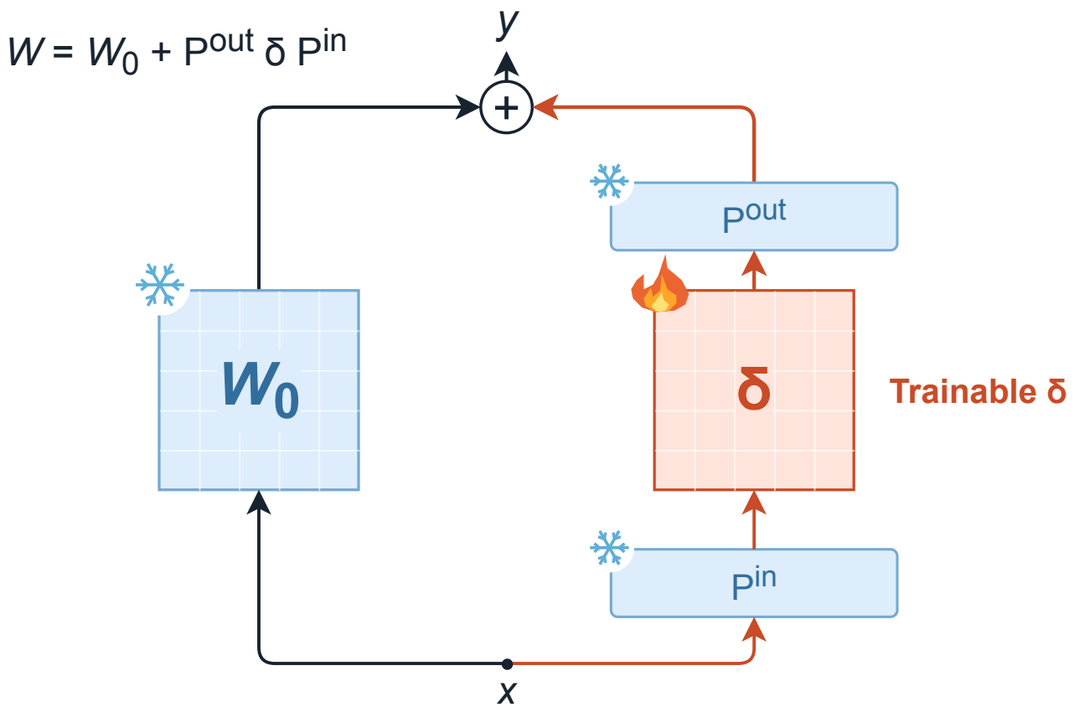


(b) CurvDelta

Low task loss · low preservation loss



(c) Train the delta, freeze the base



Dense δ (CurvDelta) · LoRA δ = *BA* (CurvLoRA)

(d) Task performance and retention

